

Vectors in $\mathbb{R}^n$

$(0, 1)$ are points in $\mathbb{R}^2$

$(-1, 2, 5)$ " " in $\mathbb{R}^3$

$(0, -1, 4, 2)$ " " in $\mathbb{R}^4$

$u = (u_1, u_2, \dots, u_n)$ in $\mathbb{R}^n$

equal of Vectors

$$\text{If } (x, 3) = (2, x+y)$$

then $x = \dots$, $y = \dots$

sol

$$x = 2 \rightarrow (1)$$

$$x + y = 3 \rightarrow (2)$$

$$2 + y = 3$$

$$y = 3 - 2$$

$$\boxed{y = 1}$$

$$\text{If } (2, -3, 4) = x(1, 1, 1) + y(1, 1, 0) + z(1, 0, 0)$$

$$\text{then } x = \dots$$

$$y = \dots$$

$$z = \dots$$

$$(2, -3, 4) = (x, x, x) + (y, y, 0) + (z, 0, 0)$$

$$(2, -3, 4) = (x+y+z, x+y, x)$$

$$x + y + z = 2$$

$$-3 = x + y$$

$$\boxed{x = 4}$$

$$-3 = 4 + y$$

$$-3 - 4 = y$$

$$\boxed{-7 = y}$$

$$4 - 7 + z = 2$$

$$-3 + z = 2$$

$$z = 2 + 3 = 5$$

$$\boxed{z = 5}$$

Vectors in $\mathbb{R}^n$

If u and v be vectors in $\mathbb{R}^n$

$$u = (u_1, u_2, u_3, \dots, u_n)$$

$$v = (v_1, v_2, v_3, \dots, v_n)$$

Dot product
النقط

$$u \cdot v = (u_1)(v_1) + (u_2)(v_2) + (u_3)(v_3) + \dots + (u_n)(v_n)$$

ملاحظة
If $u \cdot v = 0$

The vectors u and v are said to be

orthogonal if their dot product is zero

$$\text{If } u \cdot v = 0$$

$$\text{If } u = (2, -2, 3, 4) \quad v = (4, -2, 1, 2)$$

find $u \cdot v$

sol

$$u \cdot v = (2, -2, 3, 4) \cdot (4, -2, 1, 2)$$

$$= (2)(4) + (-2)(-2) + (3)(1) + (4)(2)$$

$$= 8 + 4 + 3 + 8 = 23$$

$$\text{If } u = (1, -2, 3, -4) \quad w = (5, -4, 5, 7)$$

find $u \cdot w$

sol

$$u \cdot w = 5 + 8 + 15 - 28 = 23 - 28 = -5$$

u, w orthogonal

t-

Vectors in $\mathbb{R}^n$

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$$v = (v_1, v_2, v_3, \dots, v_n)$$

Dot product
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$$= 8 + 4 + 3 + 8 = 23$$

$$\text{If } u = (1, -2, 3, -4) \quad w = (5, -4, 5, 7)$$

find $u \cdot w$

sol

$$u \cdot w = 5 + 8 + 15 - 28 = 28 - 28 = 0$$

u, w orthogonal

t-

$$v = v + u$$

$$+v)$$

$$ku + kv$$

$$u_n + v_n)$$

$$\textcircled{3} \quad 2u - 3v$$

sol

$$u + v = (3, -5, 2, 1) + (4, 1, -2, 5)$$

$$= (3+4, -5+1, 2+(-2), 1+5)$$

$$= (7, -4, 0, 6)$$

$$\textcircled{2} \quad 3u = 3(3, -5, 2, 1) = (9, -15, 6, 3)$$

$$\textcircled{3} \quad 2u - 3v = (6, -10, 4, 2) + (-12, -3, 6, -15)$$

$$= (-6, -13, 10, -13)$$

t-

Vectors in $\mathbb{R}^n$

If u and v be vectors in $\mathbb{R}^n$

$$u = (u_1, u_2, u_3, \dots, u_n)$$

$$v = (v_1, v_2, v_3, \dots, v_n)$$

$$\textcircled{1} u + v = (u_1 + v_1, u_2 + v_2, u_3 + v_3, \dots, u_n + v_n)$$

$$\textcircled{2} k u = (k u_1, k u_2, k u_3, \dots, k u_n)$$

$$\text{if } u = (3, -5, 2, 1)$$

$$v = (4, 1, -2, 5)$$

Find $\textcircled{1} u + v$

$\textcircled{2} 3u$

$$\textcircled{1} u + v = v + u$$

$$k(u + v) = k u + k v$$

$$\textcircled{3} 2u - 3v$$

sol

$$u + v = (3, -5, 2, 1) + (4, 1, -2, 5)$$

$$= (3+4, -5+1, 2+(-2), 1+5)$$

$$= (7, -4, 0, 6)$$

$$\textcircled{2} \quad (3, -5) = (9, -15, 6, 3)$$

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$$= (-6, -13, 10, -13)$$

t-

Vectors in $\mathbb{R}^n$

If u and v be vectors in $\mathbb{R}^n$

$$u = (u_1, u_2, u_3, \dots, u_n)$$

$$v = (v_1, v_2, v_3, \dots, v_n)$$

Distance in $\mathbb{R}^n$ $d(u, v)$
المسافة

$$d(u, v) = \sqrt{(u_1 - v_1)^2 + (u_2 - v_2)^2 + \dots + (u_n - v_n)^2}$$

Norm

$$\|u\| = \sqrt{u \cdot u} = \sqrt{u_1^2 + u_2^2 + u_3^2 + \dots + u_n^2}$$

$$\|u - v\| = d(u, v)$$

If $u = (1, -2, 4, 1)$ and $v = (3, 1, -5, 0)$

Find $d(u, v)$ and $\|u\|$ and $\|v\|$ and $\|u - v\|$

sol

$$d(u, v) = \sqrt{(1-3)^2 + (-2-1)^2 + (4-(-5))^2 + (1-0)^2}$$

$$= \sqrt{(-2)^2 + (-3)^2 + (9)^2 + (1)^2}$$

$$= \sqrt{4 + 9 + 81 + 1} = \sqrt{95}$$

$$\|u\| = \sqrt{(1)^2 + (-2)^2 + (4)^2 + (1)^2} = \sqrt{1 + 4 + 16 + 1} = \sqrt{22}$$

$$\|v\| = \sqrt{(3)^2 + (1)^2 + (-5)^2 + (0)^2} = \sqrt{9 + 1 + 25 + 0} = \sqrt{35}$$

$$\|u - v\| = d(u, v) = \sqrt{95}$$

t-

Vectors in $\mathbb{C}^n$

$$\text{if } u = (z_1, z_2, z_3, \dots, z_n)$$

$$v = (w_1, w_2, w_3, \dots, w_n)$$

$$u \cdot v = z_1 \bar{w}_1 + z_2 \bar{w}_2 + \dots + z_n \bar{w}_n$$

norm

$$\|u\| = \sqrt{z_1 \bar{z}_1 + z_2 \bar{z}_2 + z_3 \bar{z}_3 + \dots + z_n \bar{z}_n}$$

$$= \sqrt{|z_1|^2 + |z_2|^2 + |z_3|^2 + \dots + |z_n|^2}$$

$$i^2 = -1$$

$$z = x + iy$$

$$\bar{z} = x - iy$$

$$z \cdot \bar{z} = x^2 + y^2 = |z|^2$$

$$|z| = \sqrt{x^2 + y^2}$$

Vectors in $\mathbb{R}^n$

A vector $\mathbf{e}$ is called

a unit vector if $\|\mathbf{e}\| = 1$

$$\cos(\theta) = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}$$

$$\theta = 90^\circ \quad \text{or} \quad \theta = \frac{\pi}{2}$$

Vectors in $\mathbb{R}^n$

If u and v be vectors in $\mathbb{R}^n$

$$u = (u_1, u_2, u_3, \dots, u_n)$$

$$v = (v_1, v_2, v_3, \dots, v_n)$$

Distance in $\mathbb{R}^n$ $d(u, v)$
المسافة

$$d(u, v) = \sqrt{(u_1 - v_1)^2 + (u_2 - v_2)^2 + \dots + (u_n - v_n)^2}$$

Norm

$$\|u\| = \sqrt{u \cdot u} = \sqrt{u_1^2 + u_2^2 + u_3^2 + \dots + u_n^2}$$

$$\|u - v\| = d(u, v)$$

If $u = (1, -2, 4, 1)$ $v = (3, 1, -5, 0)$
Find $d(u, v)$ and $\|u\|$ and $\|v\|$
 $\|u - v\|$

sol

$$d(u, v) = \sqrt{(1-3)^2 + (-2-1)^2 + (4-(-5))^2 + (1-0)^2}$$

$$= \sqrt{(-2)^2 + (-3)^2 + (9)^2 + (1)^2}$$

$$= \sqrt{4 + 9 + 81 + 1} = \sqrt{95}$$

$$\|u\| = \sqrt{(1)^2 + (-2)^2 + (4)^2 + (1)^2} = \sqrt{1 + 4 + 16 + 1} = \sqrt{22}$$

$$\|v\| = \sqrt{(3)^2 + (1)^2 + (-5)^2 + (0)^2} = \sqrt{9 + 1 + 25 + 0} = \sqrt{35}$$

$$\|u - v\| = d(u, v) = \sqrt{95}$$

t-

Vectors in $\mathbb{R}^n$

If u and v be vectors in $\mathbb{R}^n$

$$u = (u_1, u_2, u_3, \dots, u_n)$$

$$v = (v_1, v_2, v_3, \dots, v_n)$$

Distance in $\mathbb{R}^n$
المسافة $d(u, v)$

$$d(u, v) = \sqrt{(u_1 - v_1)^2 + (u_2 - v_2)^2 + \dots + (u_n - v_n)^2}$$

Norm

$$\|u\| = \sqrt{u \cdot u} = \sqrt{u_1^2 + u_2^2 + u_3^2 + \dots + u_n^2}$$

$$\|u - v\| = d(u, v)$$

If $u = (1, -2, 4, 1)$ and $v = (3, 1, -5, 0)$
Find $d(u, v)$ and $\|u\|$ and $\|v\|$
and $\|u - v\|$
sol

$$d(u, v) = \sqrt{(1-3)^2 + (-2-1)^2 + (4-(-5))^2 + (1-0)^2}$$

$$= \sqrt{(-2)^2 + (-3)^2 + (9)^2 + (1)^2}$$

$$= \sqrt{4 + 9 + 81 + 1} = \sqrt{95}$$

$$\|u\| = \sqrt{(1)^2 + (-2)^2 + (4)^2 + (1)^2} = \sqrt{1 + 4 + 16 + 1} = \sqrt{22}$$

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$$\|u - v\| = d(u, v) = \sqrt{95}$$

t-

If $u = (3-2i, 4i, 1+6i)$

$v = (5+i, 2-3i, 5)$

find

① $u + v$

② $u \cdot v$

③ $4iu$

④ $\|u\|$

⑤ $(1-2i)u + (3+i)v$
موجب

$$\|u\| = \sqrt{(3-2i)(3+2i) + (4i)(-4i) + (1+6i)(1-6i)}$$

$$= \sqrt{(3-2i)(3+2i) + 4i(-4i) + (1+6i)(1-6i)}$$

$$= \sqrt{\left(\underset{9}{3^2} + \underset{4}{2^2}\right) + \underset{4}{4^2} + \left(\underset{1}{1^2} + \underset{36}{6^2}\right)}$$

$$= \sqrt{13 + 16 + 37} = \sqrt{66}$$

$$i^2 = -1$$

$$z = x + iy$$

$$\bar{z} = x - iy$$

$$z \cdot \bar{z} = x^2 + y^2 = |z|^2$$

$$|z| = \sqrt{x^2 + y^2}$$

$$(i + 24i^2) = (8 + 12i, -16, -24 + 24i)$$

If $u = (3-2i, 4i, 1+6i)$
 $v = (5+i, 2-3i, 5)$

find

① $u + v$

② $u \cdot v$

③ $4i u$

④ $\|u\|$

⑤ $(1-2i)u + (3+i)v$

① $u + v = (8-i, 2+i, 6+6i)$

② $u \cdot v = (3-2i)(\overline{5+i}) + 4i(\overline{2-3i}) + (1+6i)(\overline{5})$
 $= (3-2i)(5-i) + 4i(2+3i) + (1+6i)(5)$
 $= 15 - 3i - 10i - 2 + 8i - 12 + 5 + 30i$
 $= 6 + 25i$

③ $4i u = 4i(3-2i, 4i, 1+6i) = (12i - 8i^2, 16i^2, 4i + 24i^2) = (8+12i, -16, -24+4i)$

$i^2 = -1$

$z = x + iy$

$\bar{z} = x - iy$

$z \cdot \bar{z} = x^2 + y^2 = |z|^2$

$|z| = \sqrt{x^2 + y^2}$

$$(y, z-1) = (4, 2, 3)$$

z

①

②

③

$$\therefore x + y = 2$$

$$3 + y = 2$$

$$y = 2 - 3$$

$$y = -1$$

If u, v are orthogonal

where

$$u = (2, 3k, -4, 1, 5) \quad v = (6, -1, 3, 7, 2k)$$

find k

sol

$\therefore u, v$ orthogonal.

$$-1) = (4, 2, 3)$$

If u, v are orthogonal

where

$$u(1, k, -3) \quad v(2, -5, 4)$$

find k

sol $\therefore u, v$ orthogonal.

$$u \cdot v = 0$$

$$(1)(2) + (k)(-5) + (-3)(4) = 0$$

$$2 - 5k - 12 = 0$$

$$-5k - 10 = 0$$

$$-5k = 10 \quad \therefore -5$$

$$k = -2 \quad k = \frac{10}{-5} = -2$$

$$\text{If } \|u\| = \sqrt{39}, \quad u = (1, k, -2, 5)$$

find k

sol

$$\|u\| = \sqrt{(1)^2 + (k)^2 + (-2)^2 + (5)^2}$$

$$\sqrt{39} = \sqrt{1 + k^2 + 4 + 25}$$

$$\sqrt{39} = \sqrt{30 + k^2}$$

$$39 = 30 + k^2$$

$$39 - 30 = k^2$$

$$k = \pm 3$$

If

$$(x-y, x+y, z-1) = (4, 2, 3)$$

find x, y, z

$$x-y = 4 \rightarrow (1)$$

$$x+y = 2 \rightarrow (2)$$

$$z-1 = 3 \rightarrow (3)$$

$$z = 3+1$$

$$z = 4$$

$$(1) - (2)$$

$$2x = 6$$

$$x = 3$$

$$\therefore x+y = 2$$

$$3+y = 2$$

$$y = 2-3$$

$$y = -1$$

If u, v are orthogonal

where

$$u(1, k, -3) \quad v(2, -5, 4)$$

find k

sol $\therefore u, v$ orthogonal.

$$u \cdot v = 0$$

$$(1)(2) + (k)(-5) + (-3)(4) = 0$$

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$$k = -2 \quad k = \frac{10}{-5} = -2$$

If $\|u\| = \sqrt{39}$, $u = (1, k, -2, 5)$

find k

sol

$$\|u\| = \sqrt{(1)^2 + (k)^2 + (-2)^2 + (5)^2}$$

$$\sqrt{39} = \sqrt{1 + k^2 + 4 + 25}$$

$$\sqrt{39} = \sqrt{30 + k^2}$$

$$39 = 30 + k^2$$

$$39 - 30 = k^2$$

$$k^2 = 9$$

$$k = \pm 3$$

If

$$(x-y, x+y, z-1) = (4, 2, 3)$$

find x, y, z

$$x-y = 4 \rightarrow (1)$$

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$$y = -1$$

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$$\|u\| = \sqrt{(1)^2 + (k)^2 + (-2)^2 + (5)^2}$$

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